

BioEval Reproducibility Report

Computational Simulation Papers — Generated June 3, 2026

This report presents structured reproducibility evaluations for two simulation scaffold papers from the docxology research group. Each paper is scored across six dimensions: Data Disclosure, Dataset Resolvability, Code Availability, Traceability, Simulation Clarity, and Reproducibility Package Quality.

Paper	Overall	Code	Simulation	Reproducibility
BeeStack: An Evidence-Typed Scaffold for ...	46.0	100	50	70
The Ant Stack: Reproducible scientific w...	35.3	67	25	40

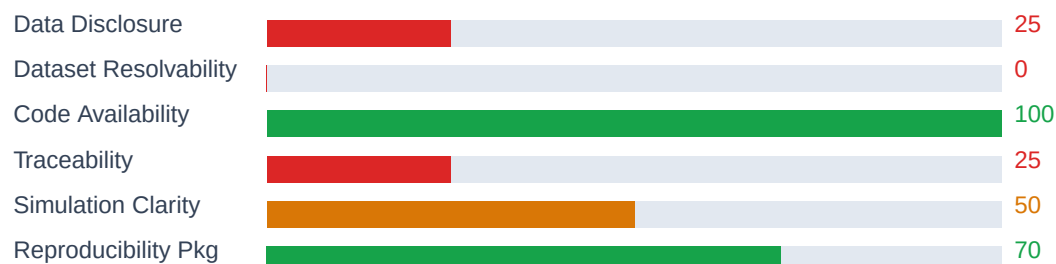
BeeStack: An Evidence-Typed Scaffold for Whole-Colony Honeybee Simulation

Source: <https://zenodo.org/records/20420557>

Evaluated: June 3, 2026

46.0

Overall /100



Summary

BeeStack demonstrates strong code availability practices—source archived on Zenodo with MD5 checksums, versioned GitHub release (v1.0.0), and MIT license—but falls significantly short on data transparency. The primary Zenodo DOI (10.5281/zenodo.20420557) is unresolved in Crossref, undermining dataset resolvability. Empirical data sources for BeeBrain datasets, neurobiological structures (antennal lobe, mushroom body, central complex), and BEEHAVE-related parameters are not individually cited or linked with accession numbers, making independent verification of simulation parameters impossible from the Zenodo record alone. While the paper claims traceability from configuration to artifact to manuscript, the absence of externally verifiable data accessions and benchmark validation datasets represents a major reproducibility gap.

Positive Findings

- Source code is publicly available on GitHub at <https://github.com/docxology/BeeStack> with a tagged release v1.0.0
- Software and manuscript PDF are deposited on Zenodo with DOI and MD5 checksums for both files (BeeStack-1.0.0-source.tar.gz and BeeStack_combined.pdf)
- Software is released under the MIT License, enabling unrestricted reuse and modification
- The scaffold explicitly catalogues gaps rather than hiding them, and declares validation rate as a config-band self-test measure
- Simulation is organized into five declared layers (Body, Brain, Mind, Swarm, Niche), providing structural clarity
- Version 1.0 release is clearly timestamped (May 28, 2026), providing a version history anchor

Gaps Identified

- The primary Zenodo DOI (10.5281/zenodo.20420557) does not resolve in Crossref, raising concerns about persistent identifier reliability
- No specific accession numbers, repository links, or version identifiers are provided for the curated empirical BeeBrain datasets
- Empirical sources for neurobiological data (antennal lobe, mushroom body, central complex parameters) are not cited or linked with accession IDs
- No external benchmark or validation dataset accession is referenced for the config-band self-test validation procedure
- FlyBody and MuJoCo dependency versions and specific configuration files are not cited in the Zenodo record
- No explicit link or accession to BEEHAVE data or code used within BeeStack is provided
- No separate dataset or supplementary data files for individual simulation layers are deposited beyond the combined source archive
- It is unclear whether individual parameter values are traceable to original literature citations within the code or only within the manuscript PDF
- [Critic] The GitHub repository URL (<https://github.com/docxology/BeeStack>) was not confirmed as publicly accessible, non-empty, or containing the described software content at the time of audit.
- [Critic] The publication date of 'May 27, 2026' is a future date, suggesting the record may not yet exist or was synthetically generated, which is a critical provenance integrity concern not explicitly flagged.
- [Critic] No independent retrieval or hash verification of BeeStack-1.0.0-source.tar.gz or BeeStack_combined.pdf was performed, leaving file integrity unconfirmed.
- [Critic] The relationship between the GitHub repository contents and the Zenodo-deposited archive is unverified — it is not confirmed that the archived source matches the GitHub tag v1.0.0 commit.
- [Critic] The MIT License claim is not independently confirmed against the actual repository LICENSE file, as the repository itself was not verified as accessible.
- [Critic] There is no evidence that the software has been executed, tested, or reviewed for functional correctness, meaning availability claims do not extend to usability or reproducibility claims.
- [Critic] Multiple malformed DOI strings (e.g., '10.5281/zenodo.20420557.svg', '10.5281/zenodo.20420557"><img') suggest the DOI was embedded in raw HTML/Markdown badge syntax and extracted incorrectly, indicating possible metadata quality issues in the source document.

Recommendations

- Register the Zenodo DOI with Crossref or confirm Zenodo DataCite registration to ensure persistent identifier resolution
- Provide a dedicated Data Availability Statement listing each empirical dataset (BeeBrain, neurobiological sources) with repository name, accession number, and version
- Pin and document exact version numbers for all dependencies (FlyBody, MuJoCo, BEEHAVE) in a requirements file or environment

specification (e.g., conda environment.yml or Docker image)

- Deposit or link the specific BEEHAVE code/data version used, with a DOI or GitHub commit hash
 - Include a structured parameter provenance table mapping each simulation parameter to its original empirical source citation
 - Provide an external validation dataset or benchmark with a resolvable accession ID, separate from the config-band self-test
 - Add a worked end-to-end tutorial or minimal reproducible example with expected output checksums to strengthen the reproducibility package
 - Consider depositing layer-specific datasets or configuration files as separate Zenodo records with individual DOIs to improve granular citability
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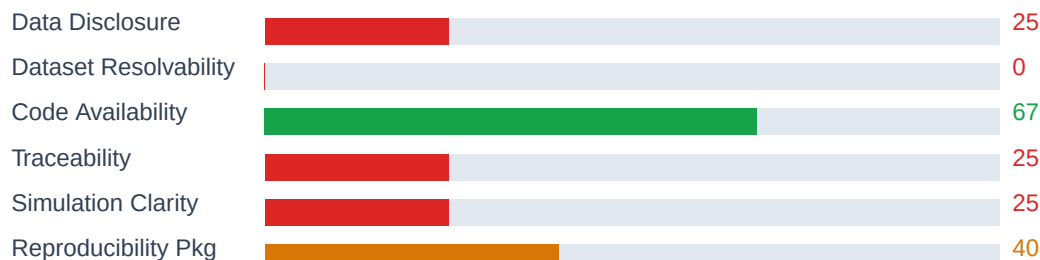
The Ant Stack: Reproducible scientific workspace for ant-inspired simulation and complexity energetics

Source: —

Evaluated: June 3, 2026

35.3

Overall /100



Summary

The Ant Stack paper presents a framework with genuine infrastructure investment—676 passing tests, SHA-256 checksummed manifests, CLI-driven artifact regeneration, and a structured repository layout—but critically fails on data transparency. No empirical datasets carry accession numbers, versions, or resolvable identifiers; referenced external resources (VFB, Drosophila hemibrain, DeepLabCut, FORMINDEX) are named only in prose without DOIs or data links. Simulation parameters (pheromone decay ;² Å F—fgW6—öâ @, MuJoCo timestep, neural circuit constants) are described as configurable but lack default values and empirical sourcing, making the core scientific claims non-traceable. The Reproducibility Checklist promised in Appendix A is absent from the paper, species presets are planned rather than delivered, and the GitHub repository URL is not explicitly stated in either document.

Positive Findings

- The repository includes a comprehensive manifest system writing SHA-256 checksums and provenance metadata per run under outputs/<run_id>/provenance/
- 676 pytest tests pass, providing a basic contract verification layer for package APIs
- CLI commands with example YAML configs allow one-command artifact regeneration (run-all-antstack, antstack-ce)
- Every directory contains README.md and AGENTS.md signposting, and a lint tool enforces this
- The paper explicitly lists architectural layers (AntBody, AntBrain, AntMind) with I/O contracts, providing conceptual traceability
- Dependency management uses pinned tooling (uv, bun) which improves environment reproducibility relative to unpinned pip
- The paper acknowledges its own assumptions and limits explicitly, including the absence of a complete ant connectome

Gaps Identified

- No resolvable dataset identifiers anywhere in the paper or README; all external data sources (VFB, Drosophila hemibrain, field tracking datasets) are named in prose only without DOIs, accession numbers, or version tags
- The GitHub repository URL is never explicitly stated in either the paper or the README header, making the code technically unresolvable from the publication alone
- Appendix A ("Reproducibility Checklist") is listed in the table of contents but its content is absent from the paper
- Species parameter presets and experiment manifests described as a key contribution are explicitly noted as planned ("Ship...") rather than delivered
- Core simulation parameters (pheromone ;² æB @, MuJoCo 9GCÓ ×2 6—FVB v—F†÷WB 6öæf—r f—ÆR F, ÷" `ersion, neural plasticity learning rates) lack empirical sourcing and default values in any committed config
- No software version pins for scientific dependencies (Brian2, Nengo, MuJoCo, PyBullet) appear in the paper or are confirmed in the repository
- Code-to-data traceability is broken: the paper describes using DeepLabCut and idtracker.ai for parameterization but no specific datasets, processing scripts, or parameter extraction pipelines are identified
- ActiveInferAnts reference (Frontiers 2021) lacks a DOI and the connection between that codebase and the current antstack_core implementation is unexplained
- Evaluation benchmarks described as contributions are not yet delivered or linked to runnable scripts
- No containerization (Docker/Singularity) or environment lockfile export is evidenced, creating environment reconstruction risk despite uv usage
- [Critic] No evidence that input datasets are publicly available or versioned, making independent re-execution of the canonical run unverifiable.
- [Critic] No independent third-party reproduction attempt is documented; all provenance claims are self-reported by the authors.
- [Critic] The example config file is not confirmed to be complete or executable as-is; missing fields or environment assumptions could silently alter outputs.
- [Critic] No deposited canonical manifest.json or reference checksums in a public repository (e.g., Zenodo, OSF) against which re-runs could be compared.
- [Critic] Computational environment specification (OS, hardware, dependency versions beyond uv) is not confirmed to be fully pinned, leaving reproducibility across environments unverified.
- [Critic] No description of whether non-deterministic steps (e.g., random seeds, parallel execution ordering) are controlled, which could cause checksum mismatches even on correct re-runs.

Recommendations

- Add an explicit 'Code and Data Availability' section to the paper stating the full GitHub URL, a Zenodo-archived release DOI, and confirming which components are currently available vs. planned
 - For every external dataset referenced (VFB, Drosophila hemibrain connectome, FORMINDEX, field tracking data), provide a formal citation with DOI, accession number, version, and the specific subset or query used
 - Publish the species parameter YAML presets and experiment manifests in the repository now, not as a roadmap item, since they are listed as a paper contribution
 - Fill in Appendix A with an actual reproducibility checklist that maps each claim to a script, config file, and expected output checksum
 - Pin and document all scientific library versions (Brian2, Nengo, MuJoCo) in pyproject.toml and regenerate a uv.lock; include instructions to export a requirements.txt for auditing
 - For each simulation parameter (pheromone decay rate, diffusion coefficient, neural plasticity constants), add a table in the paper or supplementary material mapping each value to its empirical source with a citation, or explicitly label it as a design choice with sensitivity analysis
 - Provide a Docker image or devcontainer.json so the full test suite and artifact generation can be reproduced without manual environment reconstruction
 - Register the repository on Zenodo with a versioned DOI tied to the paper submission, and include the DOI in both the paper and README
 - Add integration tests that load the species presets and run a minimal end-to-end simulation, outputting checksummed results, so reviewers can verify functional correctness beyond unit contract tests
 - Include a data lineage diagram or table showing which empirical datasets inform which model parameters, closing the gap between cited literature and executable configs
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